

Practice Quiz 05D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Contrapositive or contradiction; a universally quantified goal.

Write in full sentences. Introduce the arbitrary element and name your proof strategy.

1. Prove

$$\forall x \in \mathbf{Z} \ (8 \nmid x^3 \rightarrow 2 \nmid x).$$

Solution Let x be an arbitrary integer. We prove the contrapositive

$$2 \mid x \longrightarrow 8 \mid x^3.$$

Suppose $2 \mid x$. Then $x = 2k$ for some integer k . Therefore

$$x^3 = (2k)^3 = 8k^3.$$

Because k^3 is an integer, $8 \mid x^3$. This proves the contrapositive, so $8 \nmid x^3 \rightarrow 2 \nmid x$. Since x was arbitrary, the universally quantified statement follows. ■